MODULE 6 · LECTURE 27c

CS 6530 Applied Cryptography

Post-Quantum Cryptography Foundations

Module 6 – Post-Quantum Cryptography

Introducing Noise — Learning With Errors

What changes when a small, unknown error is added to every equation?

Exact Equations

$As = b \pmod q$

Known: (A - matrix, b - observations)

Unknown: s – secret vector

Method: Modular linear algebra.

Noisy Equations

$As + e = b \pmod q$

Known: (A - matrix, b - observations)

Unknown: (s – secret vector, e – small error vector)

Constraint: error coordinates are small.

Challenge: Recover **s** despite unknown, small errors

$$e = (e_1, e_2, \ldots, e_m)^T, \quad e_i \in [-B, B]$$

Learning With Errors (**LWE**): Given **A** and **b = As + e mod q**, recover the unknown secret **s**, where **e** contains small, unknown errors.

$A \in \mathbb{Z}_q^{m \times n}$	known matrix
$s \in \mathbb{Z}_q^{n \times 1}$	unknown secret vector
$e \in \mathbb{Z}_q^{m \times 1}$	unknown error vector
$b \in \mathbb{Z}_q^{m \times 1}$	known observation vector

$$A^{(m \times n)} \; s^{(n \times 1)} + e^{(m \times 1)} = b^{(m \times 1)} \pmod q$$

$$\begin{aligned} a_{11}s_1 + \ldots + a_{1n}s_n + e_1 &\equiv b_1 \\ a_{21}s_1 + \ldots + a_{2n}s_n + e_2 &\equiv b_2 \\ a_{31}s_1 + \ldots + a_{3n}s_n + e_3 &\equiv b_3 \pmod q \end{aligned}$$

Ignoring the errors generally gives an incorrect system. Recovering the secret requires accounting for the unknown, constrained errors.

Learning With Errors — Constructing an Instance

An LWE instance is generated by selecting a matrix, choosing a secret, adding small errors, and computing the observations.

$m = 5,\quad n = 3,\quad q = 47,\quad B = 2$

5 noisy linear equations (LWE samples) · 3 secret coordinates · modulo 47 · errors in [−2, 2]

1 Generate matrix A

$$\begin{bmatrix} 27 & 13 & 13 \\ 1 & 46 & 23 \\ 16 & 13 & 30 \\ 18 & 16 & 19 \\ 22 & 0 & 3 \end{bmatrix} = A$$

Each entry drawn uniformly from {0,...,46}; A is then fixed.

2 Choose secret s and small errors e

secret s

$$\begin{bmatrix} 2 \\ 15 \\ 12 \end{bmatrix}$$

small error e

$$\begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Illustrative values only.

3 Compute the exact product

$$\begin{bmatrix} 29 \\ 28 \\ 23 \\ 34 \\ 33 \end{bmatrix} = A\,s \bmod 47$$

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27(2)+13(15)+13(12) = 405
405 mod 47 = 29
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4 Add errors and reduce

$$\begin{bmatrix} 30 \\ 28 \\ 25 \\ 34 \\ 32 \end{bmatrix} = b$$

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29 + 1 = 30  (mod 47)
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Generating the observations (LWE samples) is straightforward when the secret and errors are known.

Learning With Errors — Recovering the Hidden Secret

The observer knows the matrix A and observations b, but does not know the secret s or errors e.

Known to the observer

$$\begin{bmatrix} 27 & 13 & 13 \\ 1 & 46 & 23 \\ 16 & 13 & 30 \\ 18 & 16 & 19 \\ 22 & 0 & 3 \end{bmatrix} = A$$

$b = (30, 28, 25, 34, 32)^T$

Unknown to the observer

$s = (s_1, s_2, s_3)^T$

$e = (e_1, \ldots, e_5)^T$

with $|e_i| \le 2$, using centered modular representatives.

The equations the observer must solve (modulo 47):

$$\begin{aligned} 27s_1 + 13s_2 + 13s_3 + e_1 &\equiv 30 \\ s_1 + 46s_2 + 23s_3 + e_2 &\equiv 28 \\ 16s_1 + 13s_2 + 30s_3 + e_3 &\equiv 25 \\ 18s_1 + 16s_2 + 19s_3 + e_4 &\equiv 34 \\ 22s_1 + 3s_3 + e_5 &\equiv 32 \end{aligned}$$

The three pairs

$s = (2, 15, 12)^T$, $e = (1, 0, 2, 0, -1)^T$

$s = (19, 34, 12)^T$, $e = (0, 2, 0, 1, 1)^T$

$s = (41, 33, 7)^T$, $e = (1, 0, -1, -2, 2)^T$

Verifying a candidate secret:

Candidate $s = (2, 15, 12)^T$ → $As \bmod 47 = (29, 28, 23, 34, 33)^T$.

$e = \text{center}_{47}(b - As)$ **$e = (1, 0, 2, 0, -1)^T$** All $|e_i| \le 2$ — the proposed secret is a valid solution.

This toy instance has three valid secret/error pairs satisfying the error bound.

Verifying a candidate secret is easy; recovering a hidden secret from noisy observations is a different computational challenge.

LWE · CRYPTOGRAPHIC APPLICATION

LWE-Based Encryption — From Public Key to Ciphertext

The receiver publishes noisy linear equations. The sender combines selected public equations and encodes a message bit.

RECEIVER — KEY GENERATION

$q = 23 \qquad n = 2 \qquad m = 4$

$$s = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \qquad e = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \qquad A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 2 & 4 \\ 4 & 2 \end{bmatrix}$$

rows 1, 3 selected

$b = As + e \pmod{23}$:

$b_1 = 1(2)+2(3)+1 = 9 \quad b_2 = 3(2)+1(3)-1 = 8 \quad b_3 = 2(2)+4(3)+0 = 16 \quad b_4 = 4(2)+2(3)+1 = 15$

$$b = (9, 8, 16, 15)^T$$

Public key: (A, b) Private key: s (e is not published)

↓ ***Public key (A, b) is sent to the sender*** ↓

SENDER — ENCRYPTION (μ = 1)

Sender's encryption inputs and outputs

r: Fresh random binary vector selecting public-key rows (1 = select, 0 = skip).
u, *v*: Two ciphertext components — **u** combines selected rows of **A**; *v* combines selected values of **b** and encodes the message bit.

$r = (1, 0, 1, 0)^T$ — selects rows 1 and 3

$u = A^T r = (\text{row } 1) + (\text{row } 3) = (1,2) + (2,4) = (3, 6)^T$

$\Delta = \lfloor 23/2 \rfloor = 11$

$v = b^T r + \mu\Delta \pmod{23} = 9 + 16 + 11 = 36 \equiv 13 \pmod{23}$

$$c = (u, v) = ((3, 6)^T, 13)$$

LWE-Based Decryption — Cancelling the Secret Component

The receiver uses the private key to cancel the linear component of the ciphertext. Only a small error remains alongside the encoded message.

CIPHERTEXT + PRIVATE KEY

$$s = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \qquad c = (u, v) = ((3, 6)^T, 13)$$

CANCELLATION

$$d = v - s^T u \pmod{23} \qquad s^T u = 2(3) + 3(6) = 24 \qquad d = 13 - 24 = -11 \equiv 12 \pmod{23}$$

Why does this recover the message?

$$\begin{aligned} d &= v - s^T u \\ &= (b^T r + \mu\Delta) - s^T A^T r \\ &= ((As+e)^T r + \mu\Delta) - s^T A^T r \\ &= e^T r + \mu\Delta \pmod{23} \end{aligned}$$

$e^T r = 1 + 0 = 1$ (selected samples: rows 1, 3)

$$d = 11 + 1 = 12 \pmod{23}$$

The large secret-dependent component has cancelled; only the encoded bit and a small error remain.

BIT DECODING

Bit 0 encodes near 0, bit 1 near 11 (mod 23); observed d = 12 is much closer to 11.

$\hat{\mu} = 1$ ***Recovered bit = original bit***

Search-LWE and Decision-LWE

Learning With Errors has two important computational formulations: recovering a hidden secret, and distinguishing noisy equations from random data.

Search-LWE

Given:

(A, b), $b = As + e \pmod{q}$

Task:

Recover the hidden secret s.

Decision-LWE

Given A and an observation vector b, distinguish between:

H_0: $b = As + e \pmod{q}$

H_1: b uniformly random in $\mathbb{Z}_q^m$

Task:

Decide whether the observations have LWE structure or are uniformly random.

Technical note: the standard Decision-LWE formulation samples the secret and error according to specified distributions. The task is to distinguish the resulting observation distribution from uniform random observations.

From LWE to Module-LWE

Module-LWE retains the noisy linear-equation structure of Learning With Errors, but its matrix and vector entries are polynomials rather than ordinary modular integers.

Ordinary LWE

$$As + e = b \pmod{q}$$

$A \in \mathbb{Z}_q^{m \times d}$, $s \in \mathbb{Z}_q^{d}$, $e, b \in \mathbb{Z}_q^{m}$

Each matrix or vector entry is an integer modulo q.

Module-LWE (MLWE)

$$As + e = t$$

where $R_q = \mathbb{Z}_q[x] / (x^n + 1)$

$A \in R_q^{k \times \ell}$, $s \in R_q^{\ell}$, $e \in R_q^{k}$, $t \in R_q^{k}$

Each entry is a polynomial in Rq; all operations use the ring arithmetic from the preceding slides.

Small-secret and small-error conditions:

$S_\eta = \{ f \in R_q : \|f\|_\infty \le \eta \}$ $\quad s \in S_{\eta 1}^{\ell}$, $\quad e \in S_{\eta 2}^{k}$

Secret: ℓ polys ≤ η1. Error: k polys ≤ η2. $\|\cdot\|_\infty$ = max |coeff|.

q — coefficient modulus
n — degree bound (deg < n)
k — rows of A (equations)
ℓ — columns of A (secret terms)
η1, η2 — bounds on secret / error coefficient magnitude

Module-LWE — A Worked Polynomial Example

Module-LWE uses the familiar noisy linear-equation structure, but its matrix and vector entries are polynomials in a ring.

$$R_7 = \mathbb{Z}_7[x] / (x^2 + 1) \qquad x^2 \equiv -1 \qquad 7 \equiv 0$$

Every polynomial in R_7 has a representative $a + bx$, with $a, b \in \mathbb{Z}_7$.

The instance: $A\,s + e = t$

$$A = \begin{bmatrix} 1+x & 2+x \\ 2-x & 1+2x \end{bmatrix} \qquad s = \begin{bmatrix} 1+x \\ 1-x \end{bmatrix} \qquad e = \begin{bmatrix} 1 \\ -x \end{bmatrix}$$

A: 2×2, s, e, t: 2×1 — every entry $\in R_7$; secret/error coefficients have centered $|\cdot| \le 1$.

First equation — t_1

$t_1 = (1+x)(1+x) + (2+x)(1-x) + 1$

Expand: $(1+x)^2 = 1+2x+x^2$

$(2+x)(1-x) = 2-x-x^2$

Combine:

$t_1 = (1+2x+x^2)+(2-x-x^2)+1$

$$t_1 = 4 + x$$

Second equation — t_2

$t_2 = (2-x)(1+x) + (1+2x)(1-x) - x$

Expand: $(2-x)(1+x) = 2+x-x^2$

$(1+2x)(1-x) = 1+x-2x^2$

Combine:

$t_2 = 3+x-3x^2$

Reduce ($x^2 \equiv -1$): $t_2 = 3+x+3$

$$t_2 = 6 + x$$

$$t = \begin{bmatrix} 4+x \\ 6+x \end{bmatrix}$$

Public: (A, t)

Hidden: (s, e)

Module-LWE — Constructing an Instance

A Module-LWE instance uses a public polynomial matrix, a small secret polynomial vector, and small errors to generate public observations.

$q = 541, \quad n = 4, \quad k = 3, \quad \ell = 2, \quad \eta_1 = 3, \quad \eta_2 = 2, \quad R_q = \mathbb{Z}_{541}[x] / (x^4 + 1)$

- A: 3 rows × 2 columns of polynomials
- e: 3 error polynomials, coefficients bounded by 2
- s: 2 secret polynomials, coefficients bounded by 3
- Every polynomial has degree < 4 after reduction

$$\begin{bmatrix} 442+502x+513x^2+15x^3 & 368+166x+37x^2+135x^3 \\ 479+532x+116x^2+41x^3 & 12+139x+385x^2+409x^3 \\ 29+394x+503x^2+389x^3 & 9+499x+92x^2+254x^3 \end{bmatrix}$$

$A \in R_q^{3\times 2}$

Coefficients sampled uniformly modulo 541.

$$\begin{bmatrix} 2-2x+x^3 \\ 3-2x-2x^2-2x^3 \end{bmatrix} \quad s \in S_3^2$$

$$\begin{bmatrix} 2-2x-x^2 \\ 1+2x+2x^2+x^3 \\ -2-x^2-2x^3 \end{bmatrix} \quad e \in S_2^3$$

$t = As + e$ in R_q^3

$$\begin{bmatrix} 30+252x+401x^2+332x^3 \\ 247+350x+259x^2+485x^3 \\ 534+234x+137x^2+443x^3 \end{bmatrix}$$

Public: (A, t)
Known — observed

Hidden: (s, e)
Unknown — the attacker's goal

Search-Module-LWE challenge: Given the public pair (A, t), recover the hidden secret vector s.

Module-LWE — Verifying the First Observation

Each observation polynomial is computed by multiplying the corresponding matrix row by the secret vector, adding the error, and reducing in the polynomial ring.

1. SELECT THE FIRST EQUATION

$$t_1 = A_{11}s_1 + A_{12}s_2 + e_1$$

$A_{11} = 442+502x+513x^2+15x^3$

$A_{12} = 368+166x+37x^2+135x^3$

$s_1 = 2-2x+x^3$

$s_2 = 3-2x-2x^2-2x^3$

$e_1 = 2-2x-x^2$

2. MULTIPLY THE POLYNOMIALS

Ordinary polynomial multiplication gives two degree-6 products, seven terms each ($A_{11}s_1$ and $A_{12}s_2$) — full expansions in the speaker notes.

3. ADD THE ERROR POLYNOMIAL

$$A_{11}s_1 + A_{12}s_2 + e_1 = 1990 - 120x - 936x^2 - 1291x^3 - 204x^4 + 169x^5 - 255x^6$$

4. REDUCE MODULO $(x^4 + 1)$

Since $x^4 = -1$, $x^5 = -x$, $x^6 = -x^2$:

$$t_1 = 2194 - 289x - 681x^2 - 1291x^3$$

Reduced polynomial powers; coefficients not yet reduced mod 541.

5. REDUCE COEFFICIENTS MODULO 541

$2194 \equiv \mathbf{30} \pmod{541}$ $-289 \equiv \mathbf{252} \pmod{541}$ $-681 \equiv \mathbf{401} \pmod{541}$ $-1291 \equiv \mathbf{332} \pmod{541}$

$$t_1 = 30 + 252x + 401x^2 + 332x^3$$

Verified: the result matches the first component of Menezes's observation vector.

Search Module-LWE and Decisional Module-LWE

Module-LWE has two related problem formulations: recovering the hidden secret, and distinguishing structured noisy observations from random observations.

Search Module-LWE (MLWE)

Given: (A, t) where $t = As + e$

with bounded secret and error coefficients.

Recover s

Decisional Module-LWE (D-MLWE)

Given: (A, z). Distinguish:

H_0: $z = As + e$

H_1: z uniformly random in R_q^k

Distinguish structured from random

In the decisional formulation, the matrix, secret and error are sampled from specified distributions; the adversary receives A and z, but not the hidden secret or error.

Security connection:

ML-KEM is based on the hardness of decisional Module-LWE.

ML-DSA uses module-lattice hardness assumptions, including D-MLWE and Module Short Integer Solution (MSIS) — the detailed MSIS construction is deferred to the ML-DSA lecture.

Why Lattices? Hardness and Cryptanalysis

Why are lattice problems relevant to the security of Module-LWE?

Why We Believe It Is Hard

Worst-case module-lattice problems
Certain lattice problems are believed difficult even for their hardest instances.

↓

Mathematical reductions
Under specified conditions, average-case MLWE also solves worst-case problems.

↓

Average-case module-lattice problems
This connection supports using randomly generated instances in cryptography.

How We Evaluate Attacks?

Module-LWE instance
Start with public noisy polynomial equations and an unknown secret.

↓

Reformulation as lattice problems
Transform the recovery task into a lattice problem for known attacks.

↓

Lattice Reduction and Attacks (e.g., LLL, BKZ)
Use techniques such as LLL and BKZ to search for exploitable lattice structure.

↓

Estimated attack cost
Estimate the computational resources required by the best applicable attacks.

Mathematical reductions provide theoretical evidence; cryptanalysis provides practical evidence for parameter selection. Neither establishes absolute security.

POST-QUANTUM CRYPTOGRAPHY · LECTURE SYNTHESIS

From Lattice Mathematics to Module-LWE

1. **Lattice Geometry** — Integer combinations of basis vectors generate lattices.
2. **Computational Hardness** — Shortest-vector and related lattice problems motivate lattice-based cryptography.
3. **Polynomial Rings** — Arithmetic reduces coefficients modulo q and polynomials modulo $x^n + 1$.

$$R_q = \mathbb{Z}_q[x] / (x^n + 1)$$

4. **Modules and Small Polynomials** — Polynomial vectors and matrices support operations with bounded secret and error coefficients.

$$R_q^k, \quad S_\eta = \{f \in R_q : |f|_\infty \le \eta\}$$

5. **Learning With Errors (LWE)** — Small unknown errors make secret recovery from modular linear equations computationally challenging for suitable parameters; LWE also supports public-key encryption.

LWE: $b = As + e \pmod q$, over $\mathbb{Z}_q$
Module-LWE: $t = As + e$, over R_q

6. **Module-LWE** — The noisy-equation structure extends to polynomial matrices and vectors, forming a foundation for post-quantum cryptographic constructions.

Next: ML-KEM — from Module-LWE-based hardness to a practical key-encapsulation mechanism.

References and Further Reading

COURSE TEXTBOOK

1. Christof Paar and Jan Pelzl — Understanding Cryptography, 2nd Edition
Chapter 12.2: Lattice-based Cryptography

FURTHER READING

2. Alfred Menezes — Kyber and Dilithium, V1: Introduction (2024)
Polynomial rings, modules, Module-LWE, and foundations for Kyber/Dilithium. cryptography101.ca/pqc/

3. Chris Peikert — A Decade of Lattice Cryptography (2016)
Formal LWE formulations, lattice problems, and hardness connections.
web.eecs.umich.edu/~cpeikert/pubs/lattice-survey.pdf

4. Oded Regev
"On Lattices, Learning with Errors, Random Linear Codes, and Cryptography," Journal of the ACM, 56(6), 2009.

Numerical toy examples and instructional diagrams adapted or developed for CS6530; they are not cryptographically secure parameter sets.